

Expt. No. 11

Write a Python program to perform following Operations on Images:

a) Addition of two images

b) Subtraction of two images.

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In [7]: import cv2
import matplotlib.pyplot as plt

# Read the images
image1 = cv2.imread('apple.jpg')
image2 = cv2.imread('baboon.jpg')

# Ensure that both images have the same dimensions
if image1.shape != image2.shape:
    print("Error: Images must have the same dimensions!")
    exit()

# Image addition
added_image = cv2.add(image1, image2)

# Image subtraction
subtracted_image = cv2.subtract(image1, image2)

# Display the images using subplot
plt.figure(figsize=(10, 10))

# Display the first image
plt.subplot(2, 2, 1)
plt.imshow(cv2.cvtColor(image1, cv2.COLOR_BGR2RGB))
plt.title("Image 1")
plt.axis('off')

# Display the second image
plt.subplot(2, 2, 2)
plt.imshow(cv2.cvtColor(image2, cv2.COLOR_BGR2RGB))
plt.title("Image 2")
plt.axis('off')

# Display the added image
plt.subplot(2, 2, 3)
plt.imshow(cv2.cvtColor(added_image, cv2.COLOR_BGR2RGB))
plt.title("Added Image")
plt.axis('off')

# Display the subtracted image
plt.subplot(2, 2, 4)
plt.imshow(cv2.cvtColor(subtracted_image, cv2.COLOR_BGR2RGB))
plt.title("Subtracted Image")
plt.axis('off')
```

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Out[7]: (-0.5, 511.5, 511.5, -0.5)
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